Elvis Arduli                No.        Date

- Knowing $x_0, x_1$ and $a$, how can we solve for $b$?
- Knowing $x_0, x_1$ and $b$, how can we solve for $a$?
- Implement basic LCG with parameters: $a=13$
  $b=10$
  $m=29$
  $X_0$ = last digit of academic ID (1)

## 1. Solving for b, given $x_0, x_1, a$:

We know that $x_1 = (a \cdot x_0 + b) \bmod m$. Let's assume $X_0=1, X_1=5, a=20, m=30$

Now we have: $5 = (20 \cdot 1 + b) \bmod 30$

Solving for b: $b = (5-20) \bmod 30$

$b = (-15) \bmod 30 = 15$. (by making the remainder positive by adding 30)

## 2. Solving for a, given $x_0, x_1, b$:

Same as before, we know: $x_1 = (a \cdot x_0 + b) \bmod m$. Lets assume that $X_0=1, X_1=5, b=3$ and $m=3$

Then: $5 = (a \cdot 1 + 3) \bmod 3$. By subtracting b from each side:

$5-3 \equiv a \bmod 3$

$2 \equiv a \bmod 3$

So $a = 2+3K$

$(2, 5, 8, 11, \ldots)$

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3 $X_0 = 1, a = 13, b = 10, m = 29$ $X_{n+1} = (a \cdot X_n + b) \bmod m$

$X_0 = 1$

$X_1 = (13 \cdot 1 + 10) \bmod 29 = 23 \bmod 29 = 23$

$X_2 = (13 \cdot 23 + 10) \bmod 29 = (299 + 10) \bmod 29 = 19$

(because $309 = 10 \cdot 29 + 19$)

$X_3 = (13 \cdot 19 + 10) \bmod 29 = 257 \bmod 29 = 25$

$\vdots$

and so on...